

PIOTR SZARAN

[Email](#) | [GitHub](#) | [Portfolio Website](#)

Full Stack Software Engineer with expertise building production applications and AI integrations across the **TypeScript**, **React**, and **Node.js** ecosystems. Experienced with **REST APIs**, **relational databases**, **Docker**, **CI/CD**, observability, **LLM** integration, and **MCP** tooling.

TECHNICAL SKILLS

Languages & Application Development

- TypeScript, JavaScript, Python, PHP, SQL, PowerShell, HTML/CSS
- React, Astro, Next.js, Node.js, Express, Hono, Tailwind CSS

Backend, Data & Integration

- REST APIs, OAuth 2.0/OIDC, JWT, MCP
- PostgreSQL, MariaDB, MySQL, SQLite, Prisma, Drizzle ORM

DevOps & Infrastructure

- Git, GitHub, GitHub Actions, Docker, Docker Compose, PM2
- Linux/Debian, Caddy, Cloudflare, Tailscale, CI/CD, Vercel

AI-Native Development Practices

- Context engineering, spec-driven development, prompt engineering, and human-in-the-loop validation
- Custom agents and skills, multi-model workflows

SELECTED ENGINEERING PROJECTS

[Pior Labs - Developer Platform](#) — Docker · PostgreSQL · OAuth 2.0/OIDC · Caddy · GitHub Actions · Terraform

- Architected and operate a self-hosted Internal Developer Platform (IDP) for independently deployed applications and shared services, centralizing OIDC SSO and OAuth 2.0 authorization, private Tailscale access, Docker networking, shared PostgreSQL infrastructure, and Caddy reverse-proxy routing and TLS.
- Standardized application bootstrapping and reduced configuration drift through a reusable React/Hono/Drizzle template, versioned design system, idempotent PostgreSQL database and role provisioning, Terraform-managed Cloudflare DNS, Docker Compose, and GitHub Actions CI/CD with repository-scoped self-hosted runners.

[Finlens - Financial Tracker](#) — React/TypeScript · Hono/Node.js · PostgreSQL/Drizzle · MCP

- Built and deployed a self-hosted finance tracker with a custom TypeScript PDF-ingestion pipeline using pdf-parse to reconstruct multi-line transactions, normalize merchant descriptions, and populate PostgreSQL-backed review workflows and spending dashboards.
- Integrated centralized SSO and a read-only MCP server, providing strict authorization boundaries while allowing external AI models to query transaction history safely.

[Personal Wedding Website](#) — Next.js · Supabase · Resend · Sentry · i18n

- Designed, launched, and operated a guest-management application for more than 200 guests, including custom RSVP, administration, bilingual support and email-broadcasting workflows.
- Maintained 100% uptime with zero recorded production errors throughout the event using Vercel serverless deployment and real-time monitoring with Sentry

Selected Client Work

- [Next Level Therapy](#): Migrated a clinic website from Squarespace to a code-owned Next.js application, reducing recurring platform costs and integrating Resend, Google Places reviews, and Setmore/Square booking workflows.
- [Council of Support for Polish Studies for the University of Toronto](#): Delivered a responsive bilingual platform using Next.js, TypeScript, Sanity CMS, and i18n, enabling client-managed English and Polish content after handover.

WORK EXPERIENCE

Judicial IT Officer

Judicial Information Technology Office, Ministry of the Attorney General, May 2025 – Present

- Provide technical support across Ontario's three courts while owning internal applications from requirements and architecture through deployment and maintenance.
- Architected a full-stack hardware asset management platform (React, TypeScript) managing end-to-end purchasing and return workflows, featuring role-based access control, transactional integrity, automated database backups, and CI/CD pipelines.
- Digitized the Network Access Request process through a custom WordPress plugin featuring conditional forms, Microsoft 365 out of office notifications, searchable records, and audit-ready exports.
- Engineered and deployed a help-desk intake application (PowerShell, WPF) across 2,600+ endpoint computers, replacing static email reporting with automated system diagnostics, input validation, file attachments, and guided ticket creation.

Administrative Coordinator with the Business Solutions, Operations & IT Unit

Animal Welfare Services, Ministry of the Solicitor General, January 2023 – May 2025

- Led the rollout of an asset management system (Hector), managing implementation, user acceptance training (UAT), and development of training materials to support adoption. Migrated over 6,000 assets from Excel to the new management platform.
- Developed Python scripts to automate data migration and inventory management, reducing errors and improving team productivity.
- Oversaw the portfolio of over 800 unique IT assets and delivered direct IT support to over 140 frontline inspectors.

EDUCATION

University of Toronto, Honours Bachelor of Science

- Statistics Major, Mathematics and Computer Science Minors